

Probeklausur T3: Elektrodynamik

Abgabe: 12:00, Freitag, 26. Juni 2026; Lösungen: Freitag, 10. Juli 2026

Useful identities

Some representations of the differential operators in curvilinear coordinates may be helpful for working on the problems.

Cylindrical coordinates

In cylindrical coordinates (ρ, φ, z) , for a scalar field ϕ , one has

$$\nabla \phi = \frac{\partial \phi}{\partial z} \mathbf{e}_z + \frac{\partial \phi}{\partial \rho} \mathbf{e}_\rho + \frac{1}{\rho} \frac{\partial \phi}{\partial \varphi} \mathbf{e}_\varphi. \quad (1)$$

For a vector field

$$\mathbf{A} = A_z \mathbf{e}_z + A_\rho \mathbf{e}_\rho + A_\varphi \mathbf{e}_\varphi$$

one has

$$\nabla \cdot \mathbf{A} = \frac{\partial A_z}{\partial z} + \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi}, \quad (2)$$

as well as

$$\begin{aligned} \nabla \times \mathbf{A} = & \frac{1}{\rho} \left(\frac{\partial}{\partial \rho} (\rho A_\varphi) - \frac{\partial A_\rho}{\partial \varphi} \right) \mathbf{e}_z \\ & + \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z} \right) \mathbf{e}_\rho + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \mathbf{e}_\varphi. \end{aligned} \quad (3)$$

The Laplace operator of a scalar field f in cylindrical coordinates is

$$\Delta f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}. \quad (4)$$

Spherical Coordinates

In spherical coordinates (r, ϑ, φ) , for a scalar field ϕ , one has

$$\nabla \phi = \frac{\partial \phi}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial \phi}{\partial \vartheta} \mathbf{e}_\vartheta + \frac{1}{r \sin \vartheta} \frac{\partial \phi}{\partial \varphi} \mathbf{e}_\varphi. \quad (5)$$

For a vector field

$$\mathbf{A} = A_r \mathbf{e}_r + A_\vartheta \mathbf{e}_\vartheta + A_\varphi \mathbf{e}_\varphi$$

one has

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \vartheta} \frac{\partial}{\partial \vartheta} (A_\vartheta \sin \vartheta) + \frac{1}{r \sin \vartheta} \frac{\partial A_\varphi}{\partial \varphi}, \quad (6)$$

as well as

$$\begin{aligned} \nabla \times \mathbf{A} = & \frac{1}{r \sin \vartheta} \left(\frac{\partial}{\partial \vartheta} (A_\varphi \sin \vartheta) - \frac{\partial A_\vartheta}{\partial \varphi} \right) \mathbf{e}_r \\ & + \frac{1}{r} \left(\frac{1}{\sin \vartheta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right) \mathbf{e}_\vartheta \\ & + \frac{1}{r} \left(\frac{\partial}{\partial r} (r A_\vartheta) - \frac{\partial A_r}{\partial \vartheta} \right) \mathbf{e}_\varphi. \end{aligned} \quad (7)$$

The Laplace operator of a scalar field f in spherical coordinates is

$$\Delta f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial f}{\partial \vartheta} \right) + \frac{1}{r^2 \sin^2 \vartheta} \frac{\partial^2 f}{\partial \varphi^2}. \quad (8)$$

Levi-Civita-Symbol

Furthermore, in Einstein summation convention, the following relations hold for the summation of two Levi-Civita symbols over one or more common indices:

$$\varepsilon^{ijk} \varepsilon^{lmk} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}, \quad (9)$$

$$\varepsilon^{ijk} \varepsilon^{njk} = 2\delta_{in}. \quad (10)$$

Problem 1 [(2+2)+(2+3)+3=12 Points] Short Questions

(1.a) Let $d \subset \mathbb{R}^2$ be a domain in the xy -plane. We consider a smooth function

$$\alpha: \mathcal{D} \rightarrow \mathbb{R}$$

as well as a smooth vector field

$$\boldsymbol{\eta}: \mathcal{D} \rightarrow \mathbb{R}^2.$$

All vector fields are regarded as three-dimensional vector fields with vanishing z -component, for example

$$\boldsymbol{\eta} = \eta_x \hat{\mathbf{x}} + \eta_y \hat{\mathbf{y}}.$$

Furthermore, let

$$\hat{\mathbf{z}} = (0, 0, 1)$$

be the unit vector perpendicular to the xy -plane.

Show the following identities:

(i) Show that

$$\nabla \cdot (\hat{\mathbf{z}} \times \nabla \alpha) = 0.$$

(ii) Additionally assume that

$$\nabla \times \boldsymbol{\eta} = 0.$$

Then show that

$$\nabla \cdot (\hat{\mathbf{z}} \times \boldsymbol{\eta}) = 0.$$

(1.b) (i) Consider the electrostatic potential in $\mathbb{R}^3$

$$\Phi(\mathbf{x}) = \Phi_0 \exp\left(-\frac{x^2 + y^2 + 2z^2}{a^2}\right),$$

where Φ_0 and $a > 0$ are constants. Determine the charge density $\rho(\mathbf{x})$ that generates this potential.

(ii) Consider the magnetostatic vector potential in $\mathbb{R}^3$

$$\mathbf{A}(\mathbf{x}) = A_0 \exp\left(-\frac{r^2 + 2z^2}{a^2}\right) \hat{\boldsymbol{\varphi}}$$

where A_0 and $a > 0$ are constants and $r = \sqrt{x^2 + y^2}$. $\hat{\boldsymbol{\varphi}}$ is the polar unit vector in cylindrical coordinates. Determine, in the Coulomb gauge, the current density $\mathbf{j}(\mathbf{x})$ that generates this vector potential.

(1.c) Let a surface S be given, defined by

$$S = \{\mathbf{x} \in \mathbb{R}^3 : z = a^2 - x^2 - y^2, z \geq 0\}. \quad (11)$$

This is the cap of an inverted paraboloid of height $a > 0$ in the half-space $z > 0$.

Now consider the vector field

$$\mathbf{v}(\mathbf{x}) = \sin^2(\phi) \mathbf{e}_r + \frac{az}{z^2 + a^2} \mathbf{e}_z + (1 - \exp(-z^2/a^2)) \mathbf{e}_\phi, \quad (12)$$

which is defined in cylindrical coordinates.

Compute the flux integral

$$\Phi = \int_S d\mathbf{A} \cdot (\nabla \times \mathbf{v}). \quad (13)$$

Solution:

(1.a) (i) We have

$$\begin{aligned}\nabla \cdot (\hat{\mathbf{z}} \times \nabla \alpha) &= \partial_u (\delta_{i3} \partial_j \alpha \varepsilon_{iju}) \\ &= \partial_u \partial_j \alpha \varepsilon_{3ju} \\ &= (\partial_y \partial_x - \partial_x \partial_y) \alpha = 0.\end{aligned}\tag{14}$$

Here Schwarz's theorem was used, since α is smooth.

(ii) Let

$$\nabla \times \boldsymbol{\eta} = \mathbf{0}.$$

Then

$$\begin{aligned}\nabla \cdot (\hat{\mathbf{z}} \times \boldsymbol{\eta}) &= \partial_u (\delta_{i3} \eta_j \varepsilon_{iju}) \\ &= \partial_u \eta_j \varepsilon_{3ju} \\ &= -(\nabla \times \boldsymbol{\eta})_3 = 0.\end{aligned}\tag{15}$$

(1.b) (i) We know

$$\Delta \Phi(\mathbf{x}) = -\frac{\rho(\mathbf{x})}{\epsilon_0}.$$

A direct calculation gives

$$\begin{aligned}\Delta \Phi(\mathbf{x}) &= \frac{2\Phi_0}{a^4} (2x^2 - a^2 + 2y^2 - a^2 + 8z^2 - 2a^2) \exp\left(-\frac{x^2 + y^2 + 2z^2}{a^2}\right) \\ &= \frac{2\Phi_0}{a^4} (2(x^2 + y^2) + 8z^2 - 4a^2) \exp\left(-\frac{x^2 + y^2 + 2z^2}{a^2}\right).\end{aligned}\tag{16}$$

Thus

$$\rho(\mathbf{x}) = -\frac{2\Phi_0\epsilon_0}{a^4} (2(x^2 + y^2) + 8z^2 - 4a^2) \exp\left(-\frac{x^2 + y^2 + 2z^2}{a^2}\right).$$

(ii) Here

$$\Delta \mathbf{A}(\mathbf{x}) = -\mu_0 \mathbf{j}(\mathbf{x}).$$

We use the identity

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times (\nabla \times \mathbf{A}).$$

For the given vector potential, we have

$$\mathbf{A}(\mathbf{x}) = A_\varphi(r, z) \hat{\boldsymbol{\varphi}}.$$

Thus

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial A_\varphi}{\partial \varphi} = 0.$$

Therefore

$$\nabla^2 \mathbf{A} = -\nabla \times (\nabla \times \mathbf{A}).$$

Concretely,

$$\nabla \times \mathbf{A} = -\frac{\partial A_\varphi}{\partial z} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial}{\partial r} (r A_\varphi) \hat{\mathbf{z}} \equiv \mathbf{B},$$

hence

$$B_r = -\frac{\partial A_\varphi}{\partial z}, \quad B_\varphi = 0, \quad B_z = \frac{1}{r} \frac{\partial}{\partial r} (r A_\varphi).$$

Therefore

$$\nabla \times \mathbf{B} = \left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r} \right) \hat{\boldsymbol{\varphi}}.$$

Combining these, one obtains

$$\begin{aligned} \nabla^2 \mathbf{A} &= -\nabla \times (\nabla \times \mathbf{A}) \\ &= -\left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r} \right) \hat{\boldsymbol{\varphi}} \\ &= \left(\frac{\partial^2 A_\varphi}{\partial z^2} + \frac{1}{r} \frac{\partial A_\varphi}{\partial r} + \frac{\partial^2 A_\varphi}{\partial r^2} - \frac{A_\varphi}{r^2} \right) \hat{\boldsymbol{\varphi}}. \end{aligned} \tag{17}$$

A direct calculation yields

$$\nabla^2 \mathbf{A} = A_\varphi \left(-\frac{8}{a^2} + \frac{4r^2}{a^4} + \frac{16z^2}{a^4} - \frac{1}{r^2} \right) \hat{\boldsymbol{\varphi}}.$$

Thus

$$\mathbf{j}(\mathbf{x}) = \frac{A_0}{\mu_0} \exp \left(-\frac{r^2 + 2z^2}{a^2} \right) \left(\frac{8}{a^2} - \frac{4r^2}{a^4} - \frac{16z^2}{a^4} + \frac{1}{r^2} \right) \hat{\boldsymbol{\varphi}}.$$

(1.c) We have

$$\partial S = \{ (x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = a^2, z = 0 \}.$$

We use Stokes' theorem and obtain

$$\Phi = \int_S \mathrm{d}\mathbf{A} \cdot (\nabla \times \mathbf{v}) = \int_{\partial S} \mathrm{d}\boldsymbol{\ell} \cdot \mathbf{v}.$$

Now

$$\mathrm{d}\boldsymbol{\ell} = \hat{\boldsymbol{\varphi}} a \mathrm{d}\varphi,$$

and ∂S is parametrized by $\varphi \in [0, 2\pi)$.

From the form of $\mathbf{v}$ on ∂S , it follows that

$$\mathbf{v}|_{\partial S} = \sin^2 \varphi \hat{\boldsymbol{\varphi}}.$$

Thus one obtains

$$\Phi = a \int_0^{2\pi} \sin^2 \varphi \mathrm{d}\varphi = \pi a.$$

Problem 2 Mirror Charges [2+6+4=12 Points]

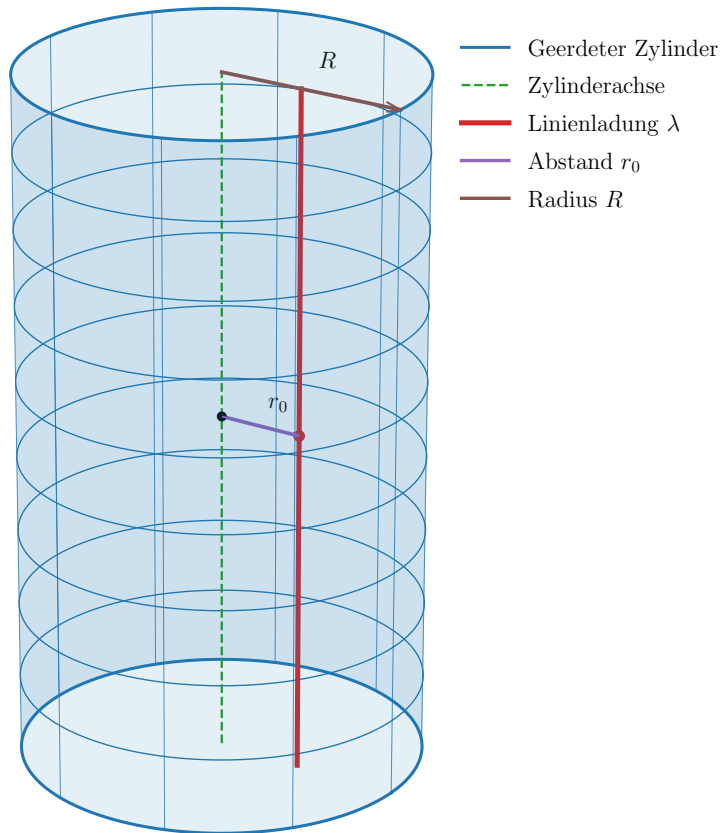


Abbildung 1: Grounded cylinder with enclosed line charge λ

An infinitely long, conducting hollow cylinder (see sketch above) of radius R is grounded. Its axis of symmetry coincides with the z -axis. Inside the cylinder there is an infinitely long line charge with constant line charge density λ . The line charge runs parallel to the z -axis and intersects the xy -plane at the point

$$\mathbf{a} = r_0 \hat{\mathbf{x}}, \quad 0 < r_0 < R.$$

The electrostatic potential is denoted by $\Phi(\mathbf{x})$.

- (2.a) State, without calculation but with a brief justification, the electrostatic potential *outside* the cylinder.
- (2.b) Now determine the electrostatic potential as well as the electric field $\mathbf{E}$ *inside* the cylinder.
- (2.c) Finally compute the induced surface charge density on the cylinder.

Hint: If you did not find a result for the electrostatic potential in part (b), continue with the following form, where β and α are free parameters and r, φ are the usual radial and angular components in cylindrical coordinates:

$$\Phi_{\text{ersatz}}(r, \varphi) = \alpha \ln \left[\frac{r^2 + \beta^2 - 2r\beta \cos \varphi}{r^2 + r_0^2 - 2rr_0 \cos \varphi} \right].$$

Solution:

(2.a) On the cylindrical surface we have

$$\Phi = 0,$$

since the cylinder is grounded. In addition, we have

$$\Phi|_{r \rightarrow \infty} = 0.$$

Since the Laplace equation must be satisfied outside the cylinder, it follows that

$$\boxed{\Phi = 0}$$

outside the cylinder.

(2.b) The position

$$\mathbf{a} = r_0 \hat{\mathbf{x}}$$

is the intersection point of the line charge with the $z = 0$ -plane.

We use the free three-dimensional Green function without boundary conditions:

$$G_0^{(3)}(\mathbf{x}; \mathbf{x}') = \frac{1}{4\pi |\mathbf{x} - \mathbf{x}'|}.$$

As an ansatz for the Green function for the problem, we choose an image line charge parallel to our physical line charge at

$$\mathbf{b} = b \hat{\mathbf{x}}.$$

Our potential is obtained by integration:

$$\Phi(\boldsymbol{\rho}) = \frac{\lambda}{\epsilon_0} \int_{-\infty}^{+\infty} dz' \frac{1}{4\pi} \left[\frac{1}{\sqrt{z'^2 + d_a^2}} - \frac{1}{\sqrt{z'^2 + d_b^2}} \right], \quad (18)$$

where

$$\boldsymbol{\rho} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}}, \quad d_a = |\boldsymbol{\rho} - \mathbf{a}|, \quad d_b = |\boldsymbol{\rho} - \mathbf{b}|.$$

We leave the integral unevaluated at first. Evaluating it gives

$$\Phi(\boldsymbol{\rho}) = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{d_b}{d_a} \right).$$

On the cylinder wall, i.e. for $r = R$, we have

$$\Phi_{\text{Rand}} = \frac{\lambda}{4\pi\epsilon_0} \ln \left(\frac{R^2 + b^2 - 2Rb \cos \varphi}{R^2 + r_0^2 - 2Rr_0 \cos \varphi} \right). \quad (19)$$

In particular, Φ_{Rand} must be independent of φ . It follows

$$b = \frac{R^2}{r_0},$$

because

$$R^2 + b^2 - 2Rb \cos \varphi = c (R^2 + r_0^2 - 2Rr_0 \cos \varphi), \quad c \in \mathbb{R}^+.$$

Comparison of coefficients gives

$$b = cr_0.$$

This yields

$$R^2 + c^2 r_0^2 = c(R^2 + r_0^2), \quad (20)$$

hence

$$(c - 1)(r_0^2 c - R^2) = 0.$$

The nontrivial solution is

$$c = \frac{R^2}{r_0^2}, \quad b = cr_0 = \frac{R^2}{r_0}.$$

Thus, concretely, on the boundary

$$\begin{aligned} \Phi_{\text{Rand}} &= \frac{\lambda}{2\pi\epsilon_0} \ln \left| \frac{R - \frac{R^2}{r_0}}{R - r_0} \right| \\ &= \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{R}{r_0} \right). \end{aligned} \quad (21)$$

Finally, this gives

$$\Phi(\boldsymbol{\rho}) = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{d_b}{d_a} \right) - \Phi_{\text{Rand}} = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{r_0}{R} \frac{|\boldsymbol{\rho} - b\hat{\mathbf{x}}|}{|\boldsymbol{\rho} - r_0\hat{\mathbf{x}}|} \right).$$

The electric field is given by the negative gradient of Φ :

$$\mathbf{E} = -\nabla\Phi = \frac{\lambda}{2\pi\epsilon_0} \left[\frac{\boldsymbol{\rho} - \mathbf{a}}{|\boldsymbol{\rho} - \mathbf{a}|^2} - \frac{\boldsymbol{\rho} - \mathbf{b}}{|\boldsymbol{\rho} - \mathbf{b}|^2} \right].$$

(2.c) On the inner cylindrical surface, the normal vector is

$$\hat{\mathbf{n}} = -\hat{\mathbf{r}}.$$

Therefore

$$\begin{aligned} \sigma(\varphi) &= \epsilon_0 \mathbf{E}(R, \varphi) \cdot (-\hat{\mathbf{r}}) \\ &= -\epsilon_0 E_r(R, \varphi) \\ &= \epsilon_0 \frac{\partial \Phi}{\partial r}(R, \varphi). \end{aligned} \quad (22)$$

Now

$$\Phi(r, \varphi) = \frac{\lambda}{4\pi\epsilon_0} \ln \left(\frac{r_0^2}{R^2} \frac{r^2 + b^2 - 2br \cos \varphi}{r^2 + r_0^2 - 2rr_0 \cos \varphi} \right).$$

It follows

$$\frac{\partial \Phi}{\partial r} = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{2r - 2b \cos \varphi}{r^2 + b^2 - 2br \cos \varphi} - \frac{2r - 2r_0 \cos \varphi}{r^2 + r_0^2 - 2rr_0 \cos \varphi} \right). \quad (23)$$

Now we set $r = R$ and $b = R^2/r_0$. This gives

$$\begin{aligned}
\left. \frac{\partial \Phi}{\partial r} \right|_{r=R} &= \frac{\lambda}{4\pi\epsilon_0} \left(\frac{2R - 2\frac{R^2}{r_0} \cos \varphi}{\frac{R^2}{r_0^2} (R^2 + r_0^2 - 2Rr_0 \cos \varphi)} - \frac{2R - 2r_0 \cos \varphi}{R^2 + r_0^2 - 2Rr_0 \cos \varphi} \right) \\
&= \frac{\lambda}{2\pi\epsilon_0} \left(\frac{\frac{r_0^2}{R} - r_0 \cos \varphi - R + r_0 \cos \varphi}{R^2 + r_0^2 - 2Rr_0 \cos \varphi} \right) \\
&= \frac{\lambda}{2\pi\epsilon_0} \left(\frac{r_0^2 - R^2}{R(R^2 + r_0^2 - 2Rr_0 \cos \varphi)} \right). \tag{24}
\end{aligned}$$

Thus, for the induced surface charge density,

$$\boxed{\sigma(R, \varphi) = \frac{\lambda}{2\pi} \left(\frac{r_0^2 - R^2}{R(R^2 + r_0^2 - 2Rr_0 \cos \varphi)} \right)}.$$

The calculation for the replacement potential from the hint proceeds identically.

Problem 3 [4+4+4=12 Points] A natural dipole field

A sphere with radius R and a uniformly charged surface with total charge Q rotates with angular velocity ω about an axis in the direction ω/ω .

(3.a) First show that the vector potential at the point $\mathbf{r}$ in the Coulomb gauge is of the form

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 Q}{(4\pi R)^2} \boldsymbol{\omega} \times \mathbf{F}(\mathbf{r})$$

and determine $\mathbf{F}(\mathbf{r})$.

(3.b) To evaluate $\mathbf{F}$, choose spherical coordinates for the integration and orient the coordinate system such that the z -axis is parallel to $\mathbf{r}$. Show that this gives

$$\mathbf{F}(\mathbf{r}) = \begin{cases} \frac{4\pi R}{3} \mathbf{r}, & r < R, \\ \frac{4\pi R^4}{3r^2} \mathbf{r}, & r > R, \end{cases} \quad r = |\mathbf{r}|.$$

(3.c) Finally show that the magnetic field outside the sphere has the form of a dipole field:

$$\mathbf{B}_{r>R}(\mathbf{r}) = \frac{3(\mathbf{m} \cdot \mathbf{r})\mathbf{r} - r^2\mathbf{m}}{r^5}.$$

Determine $\mathbf{m}$.

Solution:

(3.a) The surface charge density of the spherical shell is

$$\sigma = \frac{Q}{4\pi R^2}.$$

For a spherical shell rotating with angular velocity $\boldsymbol{\omega}$, we have

$$\mathbf{v}(\mathbf{r}') = \boldsymbol{\omega} \times \mathbf{r}'.$$

Thus the current density is

$$\mathbf{j}(\mathbf{r}') = \rho(\mathbf{r}') \mathbf{v}(\mathbf{r}') = \sigma \delta(r' - R) \boldsymbol{\omega} \times \mathbf{r}' = \frac{Q}{4\pi R^2} \delta(r' - R) \boldsymbol{\omega} \times \mathbf{r}'.$$

It follows for the vector potential

$$\begin{aligned} \mathbf{A}(\mathbf{r}) &= \frac{\mu_0}{4\pi} \int_{\mathbb{R}^3} \frac{\mathbf{j}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r' \\ &= \frac{\mu_0}{4\pi} \int_{\mathbb{R}^3} \frac{\sigma \delta(r' - R) \boldsymbol{\omega} \times \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} d^3r' \\ &= \frac{\mu_0 \sigma}{4\pi} \boldsymbol{\omega} \times \int_{\mathbb{R}^3} \frac{\delta(r' - R) \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} d^3r'. \end{aligned} \tag{25}$$

With

$$\mathbf{F}(\mathbf{r}) := \int_{\mathbb{R}^3} \frac{\delta(r' - R) \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} d^3r'$$

one therefore obtains

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 Q}{(4\pi R)^2} \boldsymbol{\omega} \times \mathbf{F}(\mathbf{r}).$$

(3.b) We first consider points on the z -axis,

$$\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix} = z \hat{\mathbf{e}}_z.$$

Then

$$\begin{aligned} \mathbf{F}(\mathbf{r}) &= \mathbf{F}(z \hat{\mathbf{e}}_z) \\ &= \int_0^\infty dr' r'^2 \int_0^{2\pi} d\varphi' \int_0^\pi d\theta' \sin \theta' \frac{\delta(r' - R) r' \hat{\mathbf{e}}_{r'}}{\sqrt{z^2 + r'^2 - 2zr' \cos \theta'}}. \end{aligned} \quad (26)$$

After carrying out the r' -integration, one obtains

$$\mathbf{F}(z \hat{\mathbf{e}}_z) = R^3 \int_0^{2\pi} d\varphi' \int_0^\pi d\theta' \frac{\sin \theta'}{\sqrt{z^2 + R^2 - 2zR \cos \theta'}} \begin{pmatrix} \cos \varphi' \sin \theta' \\ \sin \varphi' \sin \theta' \\ \cos \theta' \end{pmatrix}.$$

The integration of $\cos \varphi'$ and $\sin \varphi'$ over a full period gives 0 in each case. Thus only the z -component remains:

$$\mathbf{F}(z \hat{\mathbf{e}}_z) = \hat{\mathbf{e}}_z 2\pi R^3 \int_0^\pi \frac{\sin \theta' \cos \theta'}{\sqrt{z^2 + R^2 - 2zR \cos \theta'}} d\theta'.$$

With the substitution

$$\alpha = \cos \theta', \quad a = z^2 + R^2, \quad b = 2zR$$

one obtains

$$\mathbf{F}(z \hat{\mathbf{e}}_z) = \hat{\mathbf{e}}_z 2\pi R^3 \int_{-1}^1 \frac{\alpha}{\sqrt{a - b\alpha}} d\alpha.$$

An antiderivative gives

$$\mathbf{F}(z \hat{\mathbf{e}}_z) = -\hat{\mathbf{e}}_z \frac{4\pi R^3}{3b^2} \left[\sqrt{a - b} (2a + b) - \sqrt{a + b} (2a - b) \right]. \quad (27)$$

Thus

$$\mathbf{F}(z \hat{\mathbf{e}}_z) = \hat{\mathbf{e}}_z \begin{cases} \frac{4\pi R}{3} z, & |z| < R, \\ \frac{4\pi R^4}{3z^2}, & z \geq R. \end{cases}$$

In general, because of the rotational symmetry, the exterior case can be written as

$$\mathbf{F}(\mathbf{r}) = \frac{4\pi R^4}{3r^3} \mathbf{r}, \quad r = |\mathbf{r}| \geq R.$$

(3.c) For $r \geq R$, using the result from part (3.b), we have

$$\begin{aligned}\mathbf{A}(\mathbf{r}) &= \frac{\mu_0 Q}{(4\pi R)^2} \boldsymbol{\omega} \times \left(\frac{4\pi R^4}{3r^3} \mathbf{r} \right) \\ &= \frac{\mu_0 Q R^2}{12\pi} \frac{\boldsymbol{\omega} \times \mathbf{r}}{r^3}.\end{aligned}\tag{28}$$

Thus

$$\boxed{\mathbf{A}(\mathbf{r}) = \frac{\mu_0 Q R^2}{12\pi} \frac{\boldsymbol{\omega} \times \mathbf{r}}{r^3}, \quad r \geq R.}$$

Now

$$\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r}).$$

We use the identity

$$\nabla \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\nabla \cdot \mathbf{b}) - \mathbf{b}(\nabla \cdot \mathbf{a}) + (\mathbf{b} \cdot \nabla)\mathbf{a} - (\mathbf{a} \cdot \nabla)\mathbf{b}.$$

Since $\boldsymbol{\omega}$ is constant, it follows

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 Q R^2}{12\pi} \left[\boldsymbol{\omega} \left(\nabla \cdot \frac{\mathbf{r}}{r^3} \right) - (\boldsymbol{\omega} \cdot \nabla) \frac{\mathbf{r}}{r^3} \right].$$

For $r > R$, we have in particular $r \neq 0$, so

$$\nabla \cdot \frac{\mathbf{r}}{r^3} = 0.$$

Thus one obtains

$$\begin{aligned}\mathbf{B}(\mathbf{r}) &= -\frac{\mu_0 Q R^2}{12\pi} \left[\omega_i \partial_i \left(\frac{r_j}{r^3} \right) \hat{\mathbf{e}}_j \right] \\ &= -\frac{\mu_0 Q R^2}{12\pi} \left[\omega_i \left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5} \right) \hat{\mathbf{e}}_j \right] \\ &= \frac{\mu_0 Q R^2}{12\pi} \left(\frac{3(\boldsymbol{\omega} \cdot \mathbf{r})\mathbf{r}}{r^5} - \frac{\boldsymbol{\omega}}{r^3} \right).\end{aligned}\tag{29}$$

Thus

$$\boxed{\mathbf{B}(\mathbf{r}) = \frac{\mu_0 Q R^2}{12\pi} \frac{3(\boldsymbol{\omega} \cdot \mathbf{r})\mathbf{r} - r^2 \boldsymbol{\omega}}{r^5}, \quad r > R.}$$

With the magnetic dipole moment

$$\mathbf{m} = \frac{Q R^2}{3} \boldsymbol{\omega}$$

this can also be written as

$$\boxed{\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{3(\mathbf{m} \cdot \mathbf{r})\mathbf{r} - r^2 \mathbf{m}}{r^5}.$$

Problem 4 (Zentralübung) [4+3+3+3+3=16 Points] Rewriting the Maxwell equations

First derivatives are numerically easier to evaluate than second derivatives. Therefore the equation

$$\partial_t \mathbf{A} = -\mathbf{E} - \nabla \varphi, \quad (30)$$

is easy to integrate. In the following we use the Lorenz gauge

$$\partial_t \varphi = -c^2 \nabla \cdot \mathbf{A}. \quad (31)$$

(4.a) Let

$$\mathcal{C}(\mathbf{r}, t) := \nabla \cdot \mathbf{E} - \frac{\rho}{\varepsilon_0}. \quad (32)$$

and assume the initial conditions $\mathcal{C}(\mathbf{r}, t = 0) = 0$. Furthermore, let

$$\frac{1}{c^2} \partial_t \mathbf{E} = \nabla \times (\nabla \times \mathbf{A}) - \mu_0 \mathbf{j}, \quad (33)$$

Show that equations (30), (33) together with the Gauss condition $\mathcal{C}(\mathbf{r}, t) = 0$ for all times t imply that the fields $\mathbf{A}, \mathbf{E}$ satisfy all four Maxwell equations.

(4.b) Show that $\mathcal{C}(\mathbf{r}, t = 0) = 0$ together with the equations above and the continuity equation already implies that $\mathcal{C}(\mathbf{r}, t) = 0$ is satisfied for all times t . In other words, show that

$$\partial_t \mathcal{C}(\mathbf{r}, t) = 0. \quad (34)$$

(4.c) We now define the scalar field Γ by $\Gamma = -\frac{1}{c^2} \partial_t \varphi$. Show that equations (33) and (31) can then be rewritten as

$$\frac{1}{c^2} \partial_t \mathbf{E} = -\nabla^2 \mathbf{A} + \nabla \Gamma - \mu_0 \mathbf{j}, \quad (35)$$

$$\partial_t \Gamma = -\frac{\rho}{\varepsilon_0} - \nabla^2 \varphi. \quad (36)$$

Hint: $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$,

(4.d) Define

$$\mathcal{D} := \Gamma - \nabla \cdot \mathbf{A}. \quad (37)$$

Taking the divergence of (35) and using the continuity equation, first show

$$\partial_t \mathcal{C} = c^2 \nabla^2 \mathcal{D}. \quad (38)$$

(4.e) Differentiate (37) with respect to time to show

$$\partial_t \mathcal{D} = \mathcal{C} \quad (39)$$

Finally, combine this with the result from (d) to show that the wave equation

$$\partial_t^2 \mathcal{C} = c^2 \nabla^2 \mathcal{C}. \quad (40)$$

holds.

Solution:

- (4.a)
- First Maxwell equation $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$:
This equation is obviously satisfied if $\mathcal{C}(\mathbf{r}, t) = 0$.
 - Second Maxwell equation $\nabla \cdot \mathbf{B} = 0$:
This condition is always satisfied when the magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ is written as a curl field.
 - Third Maxwell equation $\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} \mathbf{B}$:
Here we use that

$$\mathbf{E} = -\nabla\varphi - \partial_t \mathbf{A} \quad (41)$$

$$\Rightarrow \nabla \times \mathbf{E} = -\nabla \times \nabla\varphi - \partial_t \nabla \times \mathbf{A} = -\partial_t \mathbf{B}, \quad (42)$$

since the curl of a gradient field vanishes.

- Fourth Maxwell equation $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}$:

(4.b)

$$\partial_t \mathcal{C}(\mathbf{r}, t) = \nabla \cdot \partial_t \mathbf{E} - \frac{1}{\epsilon_0} \partial_t \rho \quad (43)$$

$$= c^2 \underbrace{\nabla \cdot (\nabla \times (\nabla \times \mathbf{A}))}_{=0} - \mu_0 c^2 \nabla \cdot \mathbf{j} - \frac{1}{\epsilon_0} \partial_t \rho \quad (44)$$

$$= -\frac{1}{\epsilon_0} (\partial_t \rho + \nabla \cdot \mathbf{j}) \quad (45)$$

$$= 0. \quad (46)$$

Here, in the first step, we used (33) and in the last step the continuity equation.

(4.c) With $\Gamma = -c^2 \partial_t \varphi$ we obtain

$$\frac{1}{c^2} \partial_t \mathbf{E} = \nabla \times (\nabla \times \mathbf{A}) - \mu_0 \mathbf{j} \quad (47)$$

$$= \nabla \underbrace{(\nabla \cdot \mathbf{A})}_{-c^{-2} \partial_t \varphi = \Gamma} - \nabla^2 \mathbf{A} - \mu_0 \mathbf{j} \quad (48)$$

$$= \nabla \Gamma - \nabla^2 \mathbf{A} - \mu_0 \mathbf{j} \quad (49)$$

(4.d) Define

$$\mathcal{D} = \Gamma - \nabla \cdot \mathbf{A} \quad (50)$$

Then we take the divergence of (35)

$$\frac{1}{c^2} \nabla \cdot \partial_t \mathbf{E} = -\nabla \cdot \nabla^2 \mathbf{A} + \nabla^2 \Gamma - \mu_0 \underbrace{\nabla \cdot \mathbf{j}}_{=-\partial_t \rho} \quad (51)$$

$$\Rightarrow \partial_t (\underbrace{\nabla \cdot \mathbf{E}}_{1/\epsilon_0} - \mu_0 c^2 \rho) = c^2 \nabla^2 \underbrace{(-\nabla \cdot \mathbf{A} + \Gamma)}_{=\mathcal{D}} \quad (52)$$

$$\Rightarrow \partial_t \mathcal{C} = c^2 \nabla^2 \mathcal{D}. \quad (53)$$

(4.e)

$$\partial_t \mathcal{D} = \partial_t \Gamma - \nabla \cdot \partial_t \mathbf{A} \quad (54)$$

$$= -\frac{\rho}{\epsilon_0} - \nabla^2 \varphi + \nabla \cdot \mathbf{E} + \nabla^2 \varphi \quad (55)$$

$$= \mathcal{C} . \quad (56)$$

Substituting this into the result from (d), we obtain

$$\partial_t^2 \mathcal{C} = c^2 \nabla^2 \partial_t \mathcal{D} = c^2 \nabla^2 \mathcal{C} . \quad (57)$$